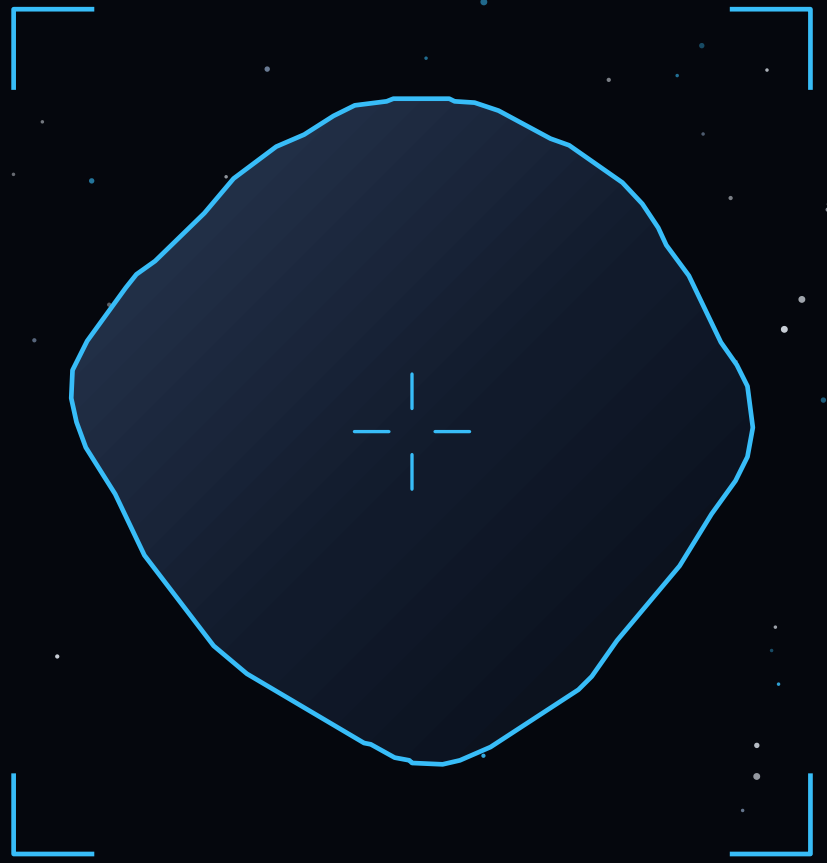
UB HACKING 2026 · DEEP-SPACE SAMPLE BAY

STARHAND

Reach into a deep-space sample bay with your bare hand.

Ahmed Sohail Butt
UNIVERSITY AT BUFFALO · NOV 7–8, 2026



BENNU · NASA SHAPE MODEL · 31 MM · 15 G

Nothing may touch Bennu

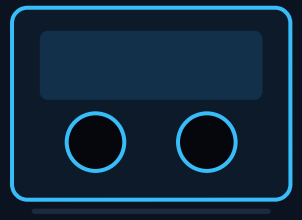
121.6 g

of asteroid Bennu, returned to Earth by NASA's OSIRIS-REx

INSIDE THE SAMPLE

5 of 5
nucleobases

14 of 20
amino acids life uses



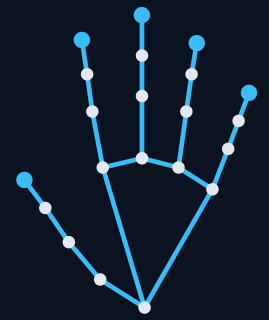
SEALED GLOVEBOXES

NASA Johnson Space Center. Only clean tools touch it.



One-way light time. Remote hands need safe autonomy.

Sense → Understand → Act

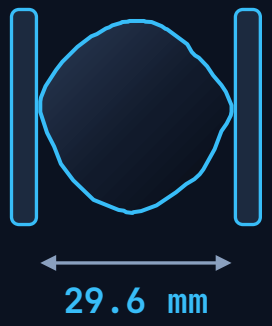


01 SENSE

Hand tracking, on-device

Fixed bay camera

~30 HZ · 21 LANDMARKS



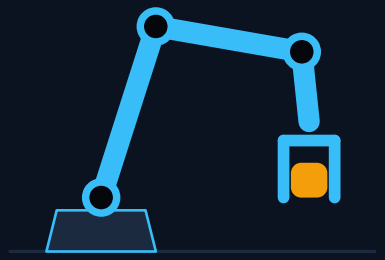
02 UNDERSTAND

Gemini designs samples

Code measures grip, height, mass

Robotics-ER checks the bay

ONE UNIFORM SCALE, OR REFUSE



03 ACT

SO-101 follows your hand

Pinch to pick, pinch to place

REACH 130–200 MM · HOVER 60 MM

STARHAND | Reach into an orbital sample-curation bay with your bare hand
CURATION BAY 2 · DEEP-SPACE STATION

SIMULATED ARM · NO PHYSICS | Leaving this machine: nothing yet | INPUT · SIMULATED KEYPAD | MET 00:07:47

Live demo

OPERATOR HAND

Simulated hand — move the pointer, press Space or click to pinch

HAND SIGNAL
No signal
Hold a gesture to command the bay

Simulated signals · hold to send — keys 1–5

1	Open palm	STOP
2	Fist, held 1.5 s	E-STOP
3	Thumb up	RESUME
4	Point up	SLOW
5	Victory	CYCLE START

Hand teleop
no hand in view — holding position

BAY STATE: RUNNING
Cycle running at full speed.
E-STOP

BAY CAMERA · LIVE
Gemini's fixed view 768²



Deck grid 1 cm · majors 5 cm

PHASE	TOOL CENTRE, MM	HOLDING	SPEED	JAW	VIEW
holding	x -1 · y -110 · z 76	nothing	×1.0	100%	1 fps

MISSION | AI proposes · code decides

COWORKER

CAPCOM · Gemini coworker
Gemini: offline | voice off

Ask CAPCOM to pick, place, check or fabricate a sample. Moves need the bay running; stop and resume stay with your hand signals.

e.g. put the Bennu sample in containn | Run

Connect voice | TALK (hold)

FABRICATOR

Fabricator — Gemini designs · code measures and decides

a 30 mm fragment of asterc | Design with Gemini

Bennu fragment | Ryugu pebble | Iron meteorite | Martian rock | Sample canister | CubeSat wheel | Ice core | Solar cell coupon | Bennu (NASA model) | Itokawa (NASA/JAXA) | Eros, full length | Kleopatra (NASA) | Antenna boom

SAMPLES ON DECK 4

- sample cube — grip 24.0 mm h 24.0 mm — box
- Bennu (NASA shape mo... — grip 29.6 mm h 31.3 mm — NASA shape model — 15 g
- Itokawa (NASA/JAXA s... — grip 21.1 mm h 17.9 mm — NASA shape model

MISSION LOG — 9 lines | Export JSON

21:25:30	VOICE	Iron meteorite fabricated. 90 grams. Scaled to fit.
21:26:02	SAFETY	TASK START - auto CYCLE START
21:26:02	VOICE	Cycle start.
21:26:02	SAFETY	CYCLE_START (auto): IDLE → RUNNING. Cycle started.
21:26:02	TELEOP	TELEOP ON — the gripper follows your hand; pinch to pick, pinch again to place

GEMINI LEDGER

CALLS	LAST LATENCY
0	—
RUNNING COST	TOKENS
$0.0000	0

Gemini proposes, code decides

GEMINI

Designs a 3D shape program

```
{"parts": […]}
```

TRUST BOUNDARY

CODE build → measure → guardrail

grip	15–45 mm
height	≤ 45 mm
mass	≤ 100 g
footprint	≤ 60 mm

SCALE TO FIT → 

REFUSE and says why, out loud

MEASURED IN THE APP · SO-101 LIMITS

✓ **Iron meteorite** 38 mm, 204 g → too heavy → ×0.76 → **28.9 mm grip, 19 mm tall, 90 g**

× **Antenna boom** refused “too tall and thin… make it at least 40 mm across, or no taller than 11 mm”

Hand signals beat the AI

WHO WINS, HIGHEST FIRST

HAND SIGNALS › PANEL › CODE GATE › AI

STOP

Open palm

SPEED → 0

E-STOP

Held fist

TELEOP OFF

RESUME

Thumbs-up

BACK TO RUNNING

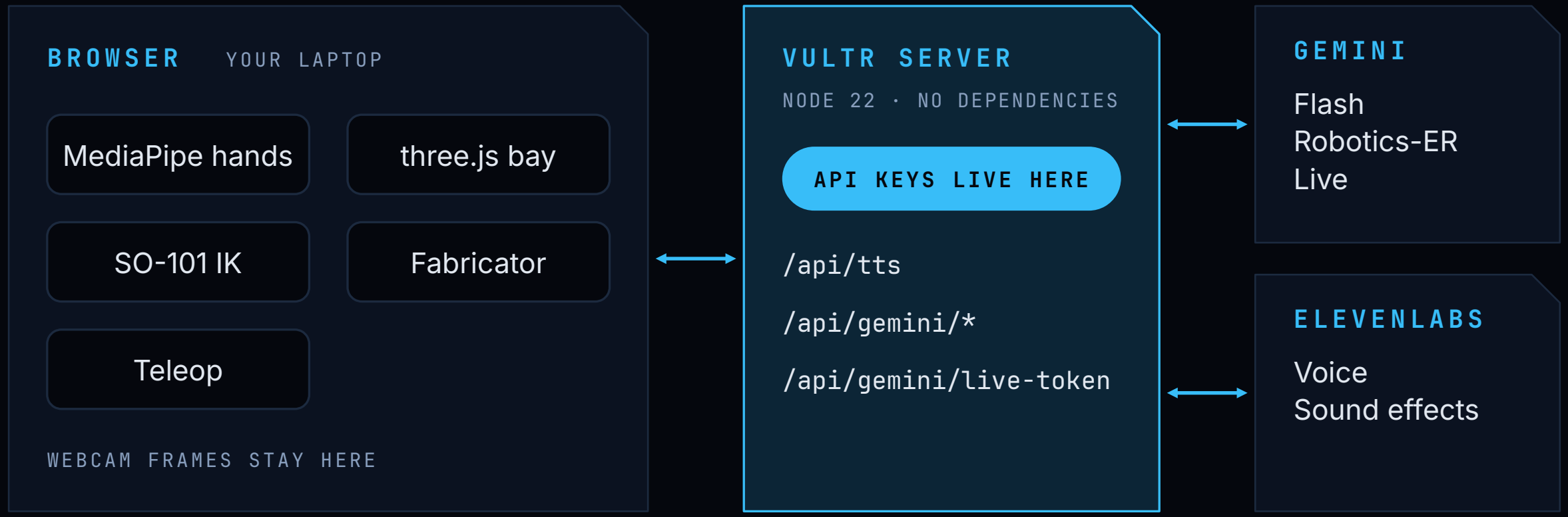
HOLD

No hand in view

DEAD-MAN: ARM WAITS

Gemini can **never** start, stop, resume or reset the bay.

No API key ever reaches the browser



Hand tracking runs on your device. Gemini Live gets a short-lived token, never the key.

What's real, what's simulated

REAL

- Hand tracking: webcam, on-device
- Voice: Gemini Live in, ElevenLabs out
- Gemini and ElevenLabs API calls
- SO-101 geometry: MuJoCo Menagerie
- NASA asteroid shape models



BENNU

ITOKAWA



EROS



KLEOPATRA

KINEMATICS CROSS-CHECKED VS MUJOCO: < 4e-8 m

SIMULATED

- Arm: kinematic, no physics
- Grasp: attaches by rule
- Mass: assumed densities

DISCLOSED IN THE APP AND README

CREDITS SO-101 model: MuJoCo Menagerie (Apache-2.0) · Asteroids and Mars: NASA 3D Resources, Itokawa NASA/JAXA, no endorsement implied · three.js, React, MediaPipe, @google/genai

AI TOOLS Claude Code wrote most of the code under my direction · Gemini and ElevenLabs run in the app · all code written at UB Hacking

What we learned, what's next

NEW SKILLS

01 On-device hand teleop

02 AI shape programs + measured guardrails

03 ElevenLabs voice design

04 Deploying on Vultr

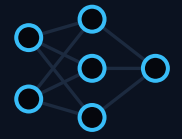
NEXT

1 Record teleop sessions as LeRobot datasets



2 Train ACT policies

ZHAO ET AL., 2023



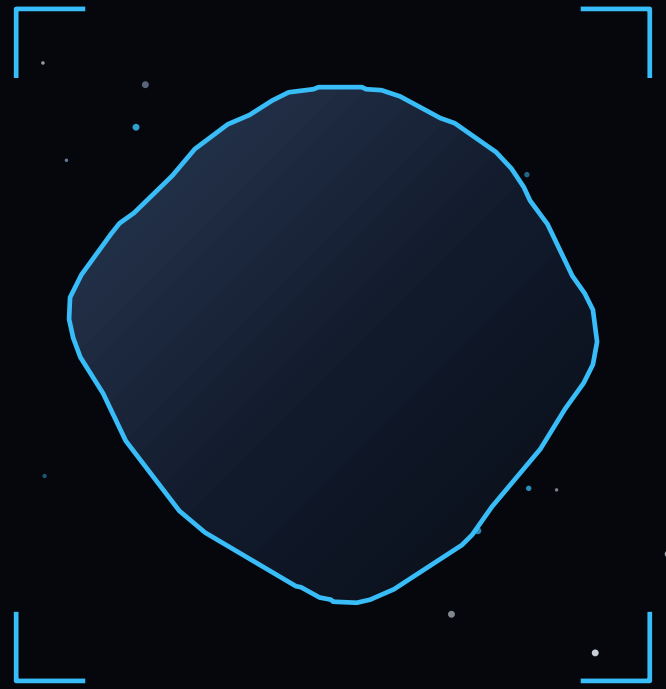
3 Run it on a real SO-101 arm



STARHAND

Thank you

Reach into deep space, safely.



REPO github.com/ahmedsohail2003/buffalo-hackathon

LIVE <live URL on Vultr>

VIDEO <YouTube link>

Ahmed Sohail Butt UB HACKING 2026

CURATION BAY 2 · DEEP-SPACE STATION

00:01:02

OPERATOR HAND

keypad

Teleop

Simulated hand — move the pointer, press Space or click to pinch

HAND SIGNAL

No signal

Hold a gesture to command the bay

Simulated signals · hold to send — keys 1–5

1	Open palm	STOP
2	Fist, held 1.5 s	E-STOP
3	Thumb up	RESUME
4	Point up	SLOW
5	Victory	CYCLE START

Hand teleop

carrying sample cube — pinch to place it in containment

BAY STATE

RUNNING

Cycle running at full speed.

E-STOP

BAY CAMERA · LIVE

Gemini's fixed view 768²

Iron meteorite

Bennu (NASA shape model)

Itokawa (NASA/JAXA shape model)

SAMPLE CONTAINMENT

Deck grid 1 cm · majors 5 cm

PHASE	TOOL CENTRE, MM	HOLDING	SPEED	JAW	VIEW
teleop	x -127 · y -127 · z …	sample cu…	×1.0	35%	3 f…

MISSION

AI proposes · code decides

COWORKER

CAPCOM · Gemini coworker

Gemini: offline · voice off

Ask CAPCOM to pick, place, check or fabricate a sample. Moves need the bay running; stop and resume stay with your hand signals.

e.g. put the Bennu sample in containn — Run

FABRICATOR

Fabricator — Gemini designs · code measures and decides

a 30 mm fragment of asterc — Design with Gemini

Bennu fragment · Ryugu pebble · Iron meteorite · Martian rock · Sample canister · CubeSat wheel · Ice core · Solar cell coupon · Bennu (NASA model) · Itokawa (NASA/JAXA) · Eros, full length

SAMPLES ON DECK

4

sample cube — held

grip 24.0 mm h 24.0 mm — box

Bennu (NASA shape mo…

grip 29.6 mm h 31.3 mm — NASA shape model

15 g

MISSION LOG

11 lines — Export JSON

21:23:27 SAFETY CYCLE_START (auto): IDLE → RUNNING. Cycle started.

21:23:2… TELEOP … again to place

21:23:53 INFO picked sample cube

GEMINI LEDGER

LAST LATENCY

RUNNING COST $0.0000

TOKENS 0

Your hand sets where. A pinch says when. No hand: the arm holds.

STARHAND | Reach into an orbital sample-curation bay with your bare hand
CURATION BAY 2 · DEEP-SPACE STATION

● SIMULATED ARM · NO PHYSICS | Leaving this machine: nothing yet | ● INPUT · SIMULATED KEYPAD | MET 00:00:38

● Fabricator

OPERATOR HAND keypad

Simulated hand — move the pointer, press Space or click to pinch

HAND SIGNAL
No signal
Hold a gesture to command the bay

Simulated signals · hold to send keys 1–5

1	Open palm	STOP
2	Fist, held 1.5 s	E-STOP
3	Thumb up	RESUME
4	Point up	SLOW
5	Victory	CYCLE START

Hand teleop

BAY STATE IDLE

Ready. The arm holds until a cycle start or a task.

E-STOP

BAY CAMERA ● LIVE

Gemini's fixed view 768²

Iron meteorite
Bennu (NASA shape model)
sample cube
Itokawa (NASA/JAXA shape model)
SAMPLE CONTAINMENT

Deck grid 1 cm · majors 5 cm

PHASE	TOOL CENTRE, MM	HOLDING	SPEED	JAW	VIEW
idle	x -1 · y -110 · z 76	nothing	×0.0	100%	1 fps

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- **Itokawa (NASA/JAXA s...** — grip 21.1 mm h 17.9 mm — NASA shape model

MISSION LOG | 5 lines | Export JSON

21:25:07 INFO Simulated hand input (?sim=1): the camera stays off
21:25:07 INFO Bay online: SO-101 arm core loaded
21:25:07 VOICE [illegible]
21:25:30 FABRICATOR FABRICATOR preset "Iron meteorite" → sample_4: scaled to 76% so the arm can lift it: 90 g, 29 mm across
21:25:30 VOICE Iron meteorite fabricated. 90 grams. Scaled to fit.

GEMINI LEDGER

CALLS	LAST LATENCY
0	—
RUNNING COST	TOKENS
$0.0000	0

Gemini designs it. Code measures it, then scales it or refuses.